Mixed benchmarks from earlier units Student code: _____ Calculator: yes

1. A probability is given for each event.

Describe each event as likely, unlikely or neither.

Check one box in each row.

	Likely	Unlikely	Neither likely nor unlikely
The probability that a tossed coin lands on heads is 50%.	☐	☐	☐
The probability that it snows in Orlando, Florida, this winter is 0.01.	☐	☐	☐
The probability that a thunderstorm is within 5 miles of your house tomorrow is $\frac{7}{10}$.	☐	☐	☐

2. Evaluate. Write your answer in simplest form.

$$\frac{1}{4} - \frac{3}{4}$$

3. A group of 4 friends owes \$148.80 for a group gift. They split the amount owed equally, and each friend's share is recorded as a negative balance. Then Maya pays $\frac{5}{6}$ of her share.

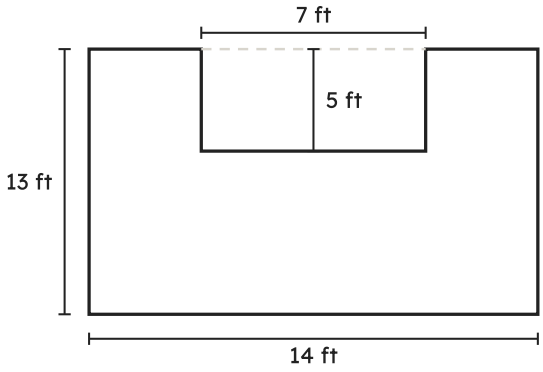
What is Maya's balance now?

Turn over →

Mixed benchmarks from earlier units Student code: _____

4. A patio is shaped like the figure shown: a 14 ft by 13 ft rectangle with a 7 ft by 5 ft rectangular notch removed from the middle of one edge. The dashed lines show the part that was removed.

What is the area of the patio, in square feet?



Not drawn to scale

5. The circumference of a round pool is 125.66 feet.

What is the diameter of the round pool? Round your answer to the nearest hundredth. Use the π key on your calculator.

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — BACKSTOP-Q2

Given when no new check is ready by Thursday. Logged as a real check (Backstop = y in CHECKS).

1. MA.7.DP.2.2 Regular **The probability that a tossed coin lands on heads is 50%. → Neither likely nor unlikely; The probability that it snows in Orlando, Florida, this winter is 0.01. → Unlikely; The probability that a thunderstorm is within 5 miles of your house tomorrow is $\frac{7}{10}$. → Likely**

Write each probability as a decimal and compare it with 0.5: an event with a probability greater than 0.5 is likely, less than 0.5 is unlikely, and exactly 0.5 is neither.

$50\% = 0.5$: exactly 0.5, so neither.

0.01: less than 0.5, so unlikely.

$\frac{7}{10} = 0.7$: greater than 0.5, so likely.

Scoring: 1 point only if every match is correct; anything else earns 0.

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2. MA.7.NSO.2.2 Regular $-\frac{1}{2}$

Both fractions already have the denominator 4.

$\frac{1}{4} - \frac{3}{4} = -\frac{2}{4}$.

Simplify: $-\frac{1}{2}$.

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3. MA.7.NSO.2.3 Regular **−$6.20**

Each share: $-\$148.80 \div 4 = -\$37.20$.

Paying $\frac{5}{6}$ of the share leaves $1 - \frac{5}{6} = \frac{1}{6}$ of it: $\frac{1}{6} \times (-\$37.20) = -\$6.20$.

Maya's balance is −$6.20.

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4. MA.7.GR.1.2 Regular $147\,\text{ft}^2$

Whole rectangle: $14 \times 13 = 182\,\text{ft}^2$.

Removed notch: $7 \times 5 = 35\,\text{ft}^2$.

Area: $182 - 35 = 147\,\text{ft}^2$.

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5. MA.7.GR.1.3 Regular **40 feet (40.00 also accepted)**

$C = \pi d$, so $d = C \div \pi$.

With the π key: $125.66 \div \pi \approx 40.00$ feet.
